Instructions — Lab 4

Before you begin

- Use Blender and work only on a copy of the starter file.
- The supplied sample files originate from Blender 2.91. Open a copy in current Blender and use Save As before editing.
- Preserve the original input and save incremental versions.
- Capture evidence at the points named below.

Step-by-step

1. Preserve the starter and create a defect-log table.
2. Apply a diagnostic material and enable wireframe/face orientation overlays.
3. Check dimensions, transforms and object scale.
4. Select non-manifold geometry and classify each finding.
5. Merge only confirmed duplicate vertices by a controlled distance.
6. Recalculate normals outside and inspect exceptions manually.
7. Repair internal faces, holes or accidental stacked extrusions locally.
8. Review edge flow around fin roots and the tail bend.
9. Reduce unnecessary density without changing the approved silhouette.
10. Pose or bend the tail to test deformation.
11. Rerun the original checks and record pass/fail results.
12. Save the repaired file and update the evidence presentation.

Expected result

A repaired, review-ready mesh with a defect log and before/after evidence.

Evidence to include

- Baseline wireframe
- Face-orientation view
- Defect log
- Deformation test
- Final statistics

Troubleshooting

- If a command or tool behaves differently, record the Blender version and compare the current Blender manual.
- If a modifier or bevel looks inconsistent, verify object scale before changing geometry.
- If an export looks wrong, re-import it into a clean file and compare dimensions, normals, hierarchy and materials.
- Never overwrite the preserved starter or baseline while diagnosing.